

CAPSTONE PROJECT REPORT

Report 4 – Software Design Document

– Ho Chi Minh, October 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
20/10/26	A	Project team	Prepared DroneOps Intelligence Software Design Document for Review 2.

*A - Added M - Modified D - Deleted

II. Software Design Document

1. System Design

1.1 System Architecture

DroneOps Intelligence uses a Next.js/React Web application, an ASP.NET Core API, application services, EF Core/Npgsql persistence and PostgreSQL. A telemetry gateway supplies normalized MAVLink-derived samples, and SSE delivers authorized live updates. Backend adapters provide media storage, email and AI integration. Low-level UAV flight control remains external.

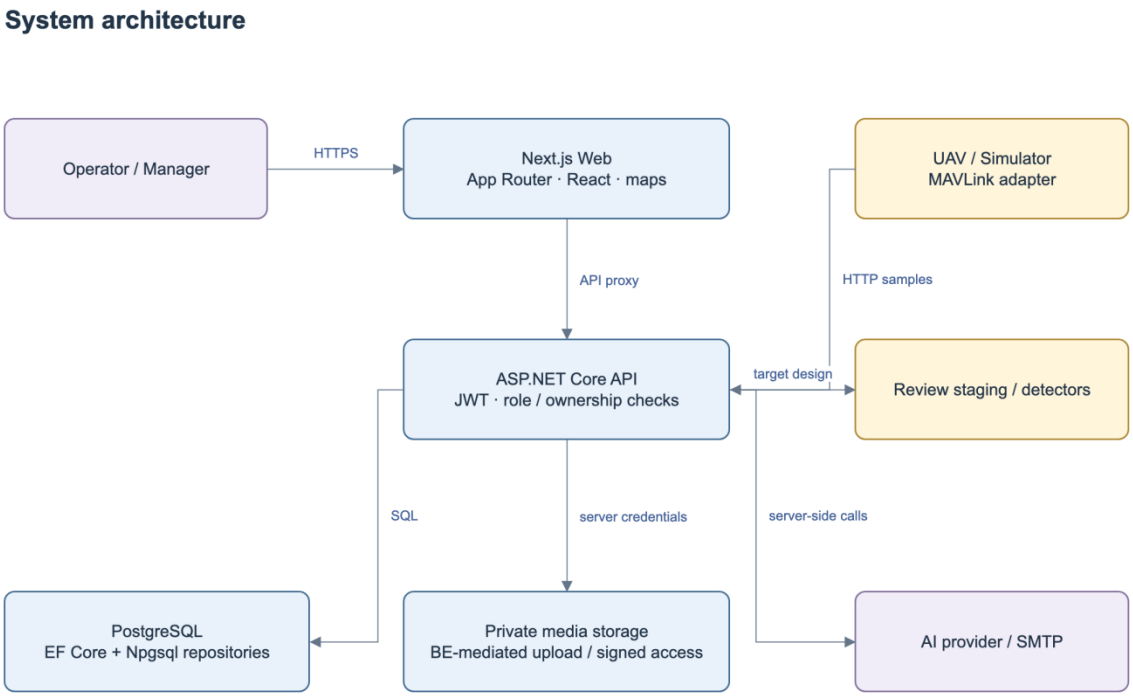


Figure 1. System Architecture.

No	Component	Description
1	Next.js Web + proxy	Role-specific pages, client validation, map, same-origin forwarding. Backend remains authoritative for authorization and state.
2	API + Application	JWT/canonical roles, DTO validation, business services and outcome errors. Controllers delegate business operations to services; the composition root configures persistence and external adapters.
3	Persistence + Domain	EF entities/repositories plus Npgsql stores. Application depends on repository interfaces in Persistence and entities in Domain.
4	PostgreSQL / media	Transactional domain records; proposed request/evidence storage. Private user media uses protected backend read/upload and short-lived access; no public server secret.

5	Integration gateway / SSE	Normalized sample HTTP endpoint exists. Raw MAVLink adapter and background loss timer remain gaps; latest snapshot and SSE support frontend monitoring.
6	External AI / email	Configured outbound calls with bounded timeout, provenance and error handling. AI advice cannot approve missions.

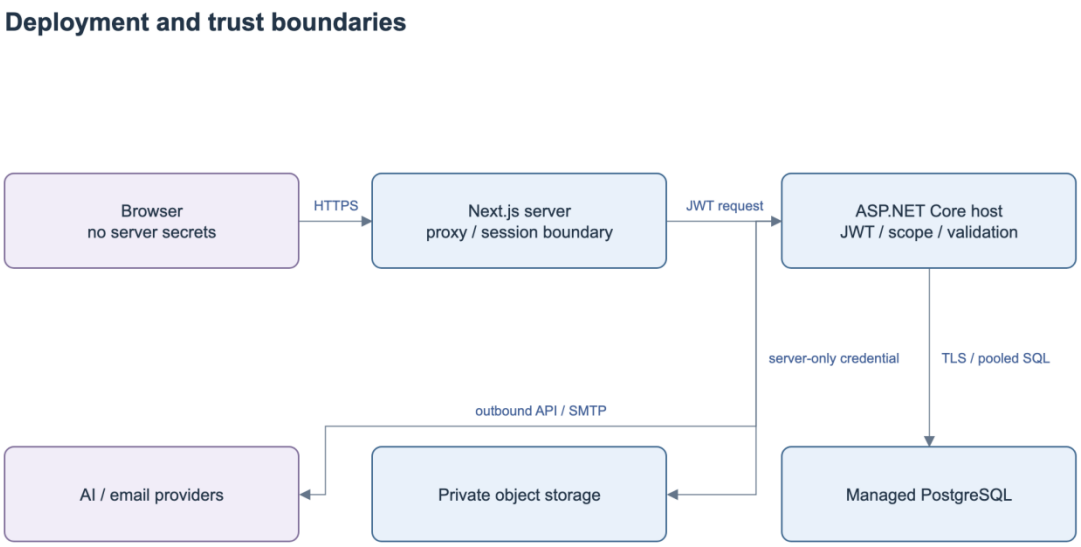


Figure 2. Deployment and Trust Boundaries.

1.2 Package Diagram

The backend packages separate API endpoints, application services, persistence and domain entities. The Web subsystem groups route components, shared clients and API forwarding.

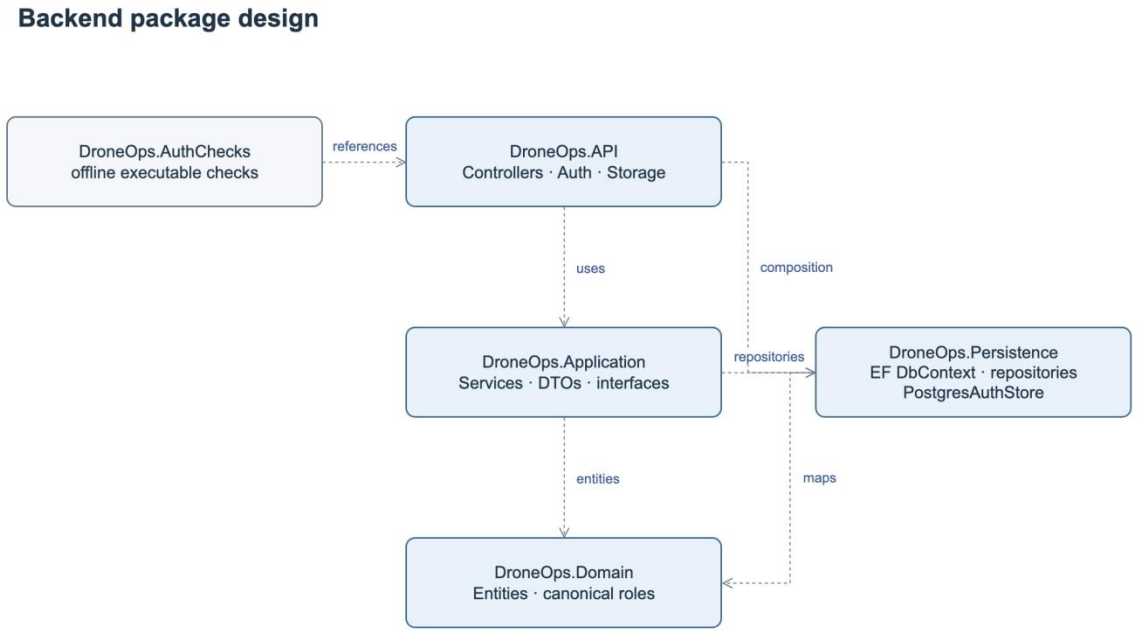


Figure 3. Backend Package Diagram.

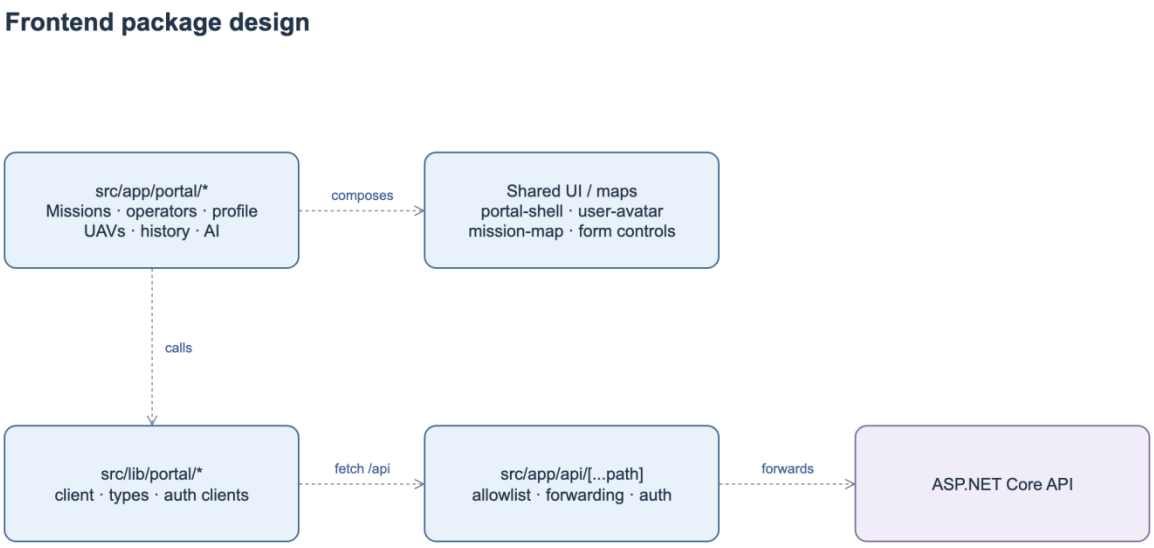


Figure 4. Web Package Diagram.

Package Descriptions

No	Package	Description
1	DroneOps.API	Controllers/{Missions,Integrations,Incidents,Users,PostFlight}; authentication middleware, Storage helpers, composition root.
2	DroneOps.Application	Services/{Missions,Telemetry,Incidents,PostFlight,AI,Auth}; request/response DTOs and service interfaces.

3	DroneOps.Persistence	AppDbContext, repositories and interfaces; PostgresAuthStore; Npgsql pooling.
4	DroneOps.Domain	Mission, Waypoint, UAV, User/Pilot, TelemetryRecord, Incident, MissionApproval, PostFlightReport and related records.
5	DroneOps.AuthChecks	Executable offline authorization/validation checks; not a full database or end-to-end test suite.
6	fe/src/app/portal	Mission/UAV/operator/profile/geofence/history/AI pages and shared shell/maps; monitoring/incident business screens still need integration.
7	fe/src/lib + API proxy	Shared DTOs/clients and server forwarding; review allowlist and streaming timeout/buffering before realtime integration.

2. Database Design

The planning model stores the current mission and proposed updates awaiting review. Telemetry, incidents, review decisions, reports and assessments retain their mission relationships. Database keys and attributes are specified below.

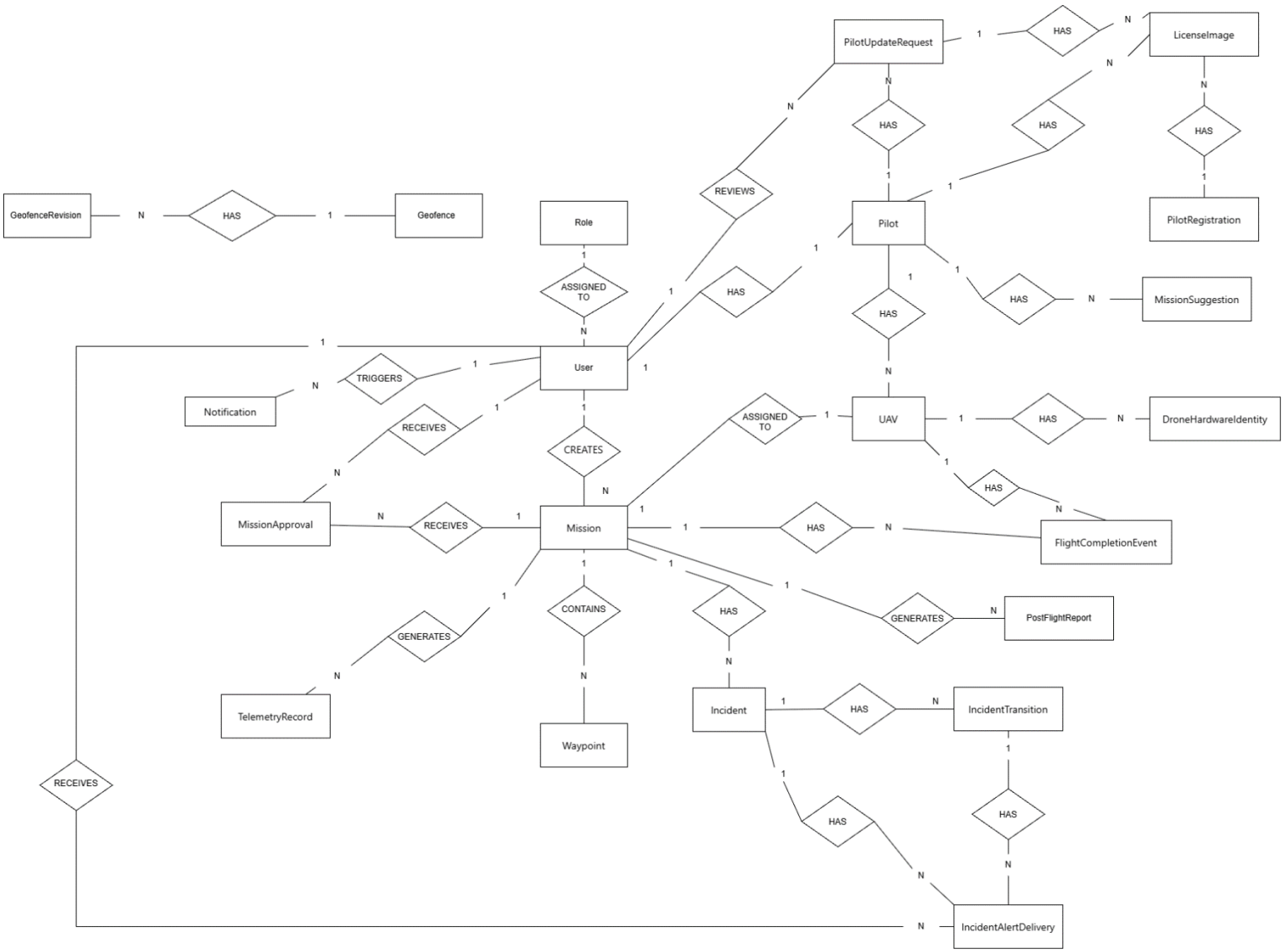


Figure 5. Entity Relationship Diagram.

Table Descriptions

No.	Entity	Domain	Description
1	Role	Identity and access	Defines an authorization category assigned to user accounts and used to control access to system capabilities.
2	User	Identity and access	Represents an authenticated system account and the actor behind mission, approval, review, and notification activities.
3	Pilot	Identity and access	Stores the operational pilot profile associated with a user and links that profile to aircraft, evidence, requests, and suggestions.
4	PilotRegistration	Identity and access	Represents a pilot onboarding or registration submission together with its supporting license evidence.

No.	Entity	Domain	Description
5	PilotUpdateRequest	Identity and access	Captures a controlled request to update pilot information and supports review and attachment of evidence before changes are accepted.
6	LicenseImage	Identity and access	Stores or references license-document imagery used as evidence for a pilot profile, registration, or profile update request.
7	UAV	Fleet and airspace	Represents a pilot-registered aircraft managed by the platform, including its registering pilot, Manager-reviewed operational status, mission allocations, hardware identities and completion events. Managed fleet means registered platform records, not ownership by the platform.
8	DroneHardwareIdentity	Fleet and airspace	Represents a hardware-level identity associated with a UAV, such as a serial, device, controller, or communication identity.
9	Geofence	Fleet and airspace	Represents a logical operational or restricted area whose definition can evolve through versioned revisions.
10	GeofenceRevision	Fleet and airspace	Represents one historical or current revision of a geofence, preserving changes to geometry or operational rules over time.
11	Mission	Mission lifecycle	Acts as the central operational record for a planned flight and connects its creator, aircraft, route, approvals, telemetry, incidents, completion events, and reports.
12	Waypoint	Mission lifecycle	Represents an ordered route point contained by a mission and used to define the intended flight path.
13	MissionApproval	Mission lifecycle	Records an approval review received by a mission and associated with the user responsible for the review activity.
14	MissionSuggestion	Mission lifecycle	Represents a mission-related recommendation or planning suggestion associated with a pilot.
15	TelemetryRecord	Mission lifecycle	Represents telemetry generated during a mission, allowing operational and post-flight analysis of aircraft state.
16	FlightCompletionEvent	Mission lifecycle	Records a completion-related lifecycle event connected to both the mission and the UAV that performed the flight.
17	PostFlightReport	Mission lifecycle	Represents a report generated for a mission after flight activity, consolidating outcome and supporting evidence.
18	Incident	Incidents and notifications	Represents an operational anomaly or safety event linked to a mission and its subsequent lifecycle and alert deliveries.
19	IncidentTransition	Incidents and notifications	Records a change in an incident's lifecycle state so that status history and operational actions remain traceable.
20	IncidentAlertDelivery	Incidents and notifications	Tracks delivery of an incident alert to a user, with context from the incident and its transition history.
21	Notification	Incidents and notifications	Represents a user-facing system notification. The source diagram labels its relationship to User as TRIGGERS and should be reviewed for intended direction.

The approval transaction checks the mission, request identity, record scope and current validation evidence before applying the accepted metadata and ordered waypoints. It records the decision and closes the request in the same transaction. Any failure rolls back the complete operation. Rejection updates only

the request and decision, preserving current mission content and prior approval. A stale or conflicting request returns 409. Notification delivery follows the committed decision.

Indexes support ordered mission waypoints, pending requests, latest mission telemetry, incident status and recipient alerts. Ingestion uses an idempotency key, while decisions use a unique request identity and terminal-state check. Backend authorization protects mission, history, media and streaming operations.

3. Detailed Design

3.1 Mission Planning and Approval

This feature covers mission submission, pre-flight validation, proposed updates and Manager decisions. Approval applies accepted content to the same mission; rejection preserves current content and any existing approval.

3.1.1 Class Diagram

The classes below separate mission editing, proposed content, review decisions and the approval transaction.

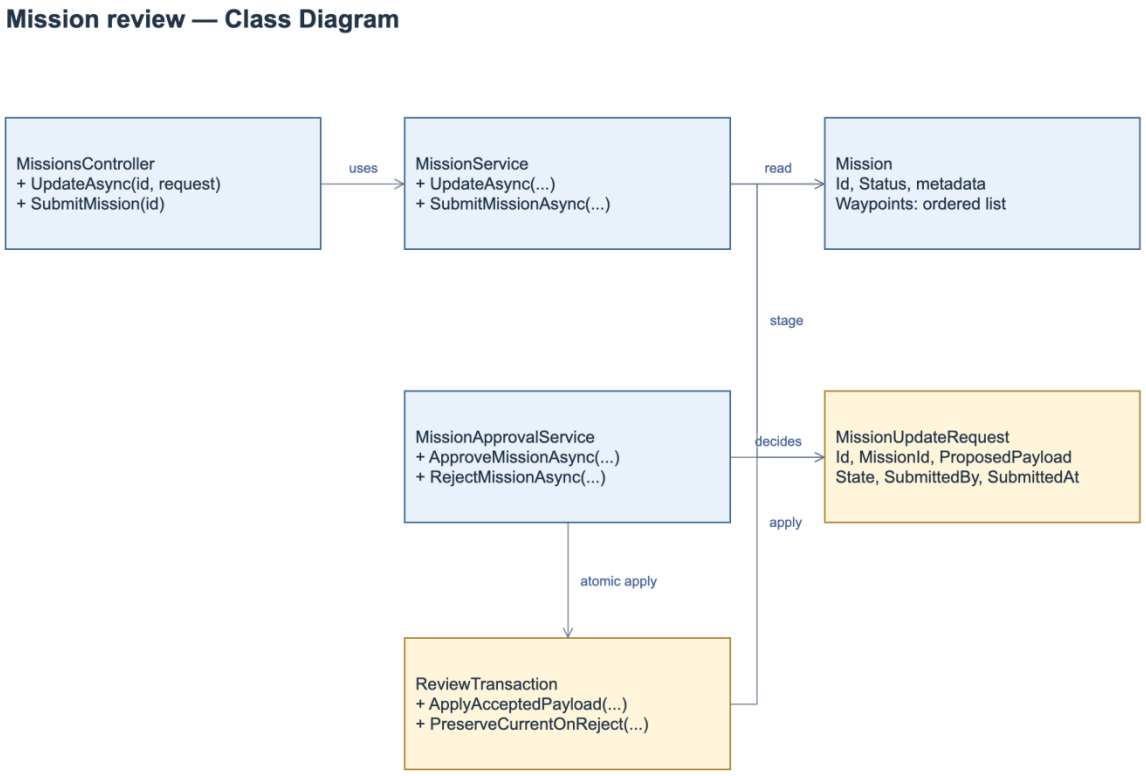


Figure 6. Mission Planning and Approval Class Diagram.

Class Specifications

No	Class / Operation	Specification

1	MissionService.UpdateAsync	Input mission ID, authenticated operator and metadata/route. Authorize scope/state; reject malformed geometry; persist separate review request; run mandatory validation; return request ID/current mission summary. Current in-place update must be replaced.
2	MissionUpdateRequest	Owns proposed payload, author/time and review state, not a mission version. Current content remains outside payload and authoritative.
3	MissionApprovalService	Approve/reject exact request, require current evidence; commit accepted data atomically or reject without content mutation. Repeated/stale decisions return current outcome or conflict.
4	Review transaction / validation	One DB unit of work, request identity, input fingerprint and restriction revision. Concurrent edits cannot reuse validation or overwrite a newer accepted proposal.

3.1.2 Submit a Mission Update

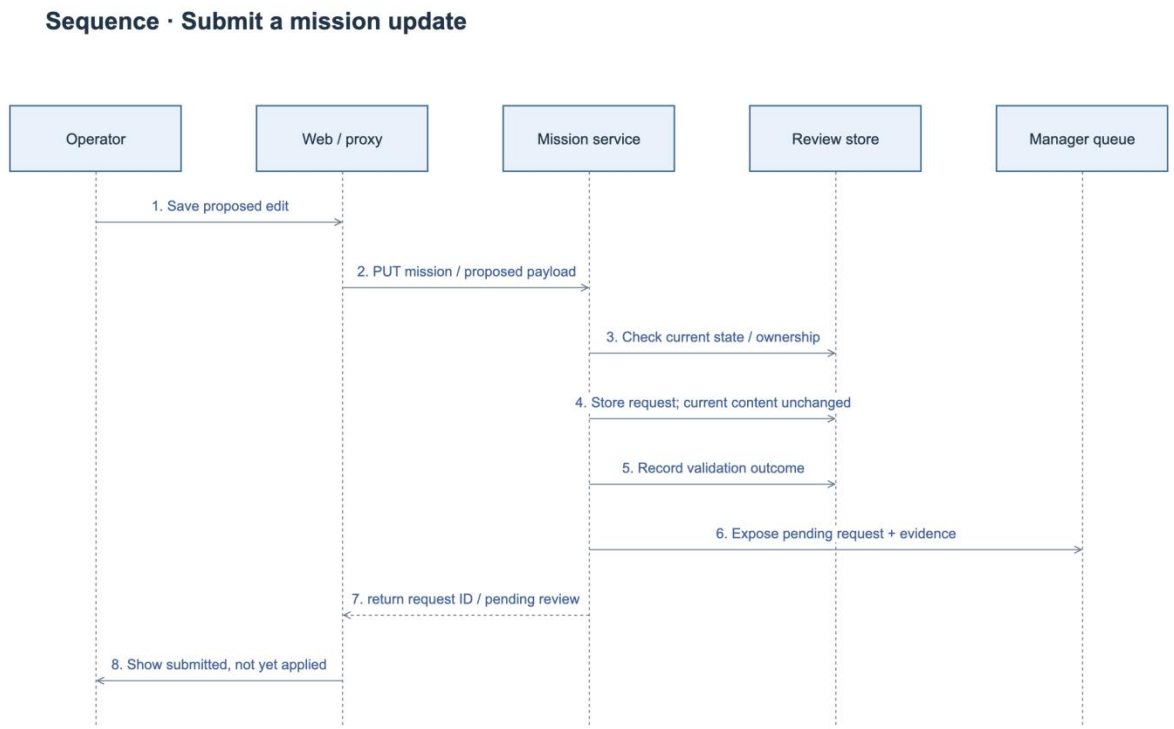


Figure 7. Submit Mission Update Sequence Diagram.

3.1.3 Approve a Mission Update

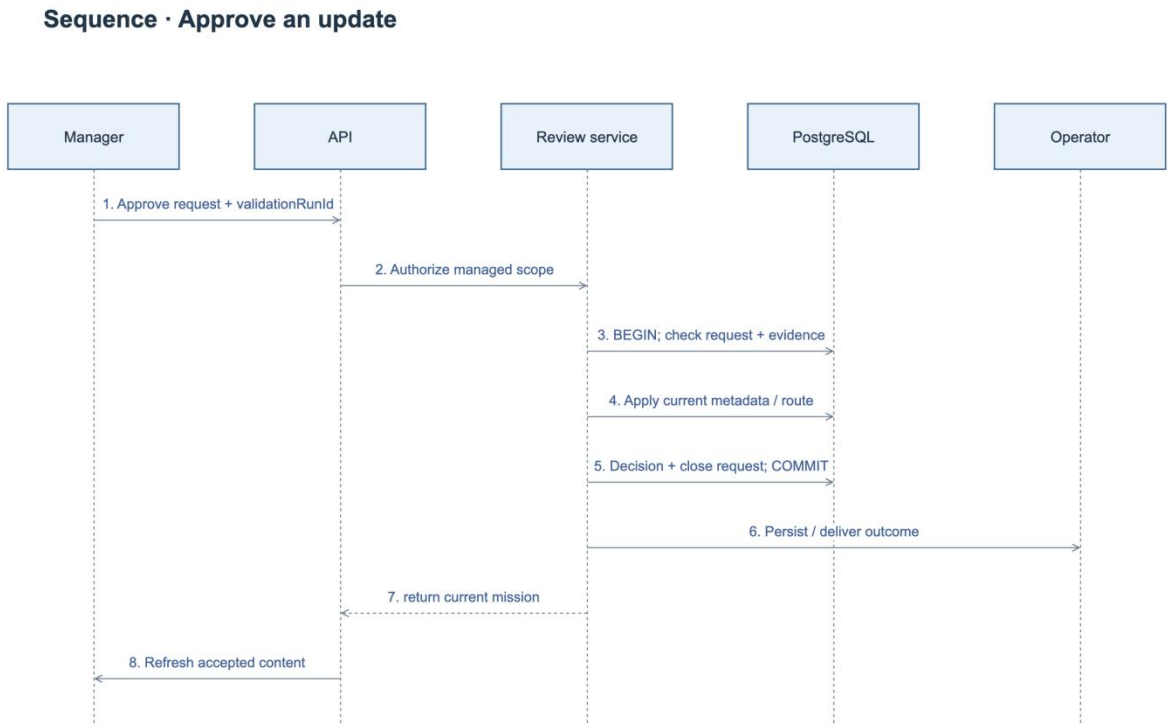


Figure 8. Approve Mission Update Sequence Diagram.

3.1.4 Reject a Mission Update

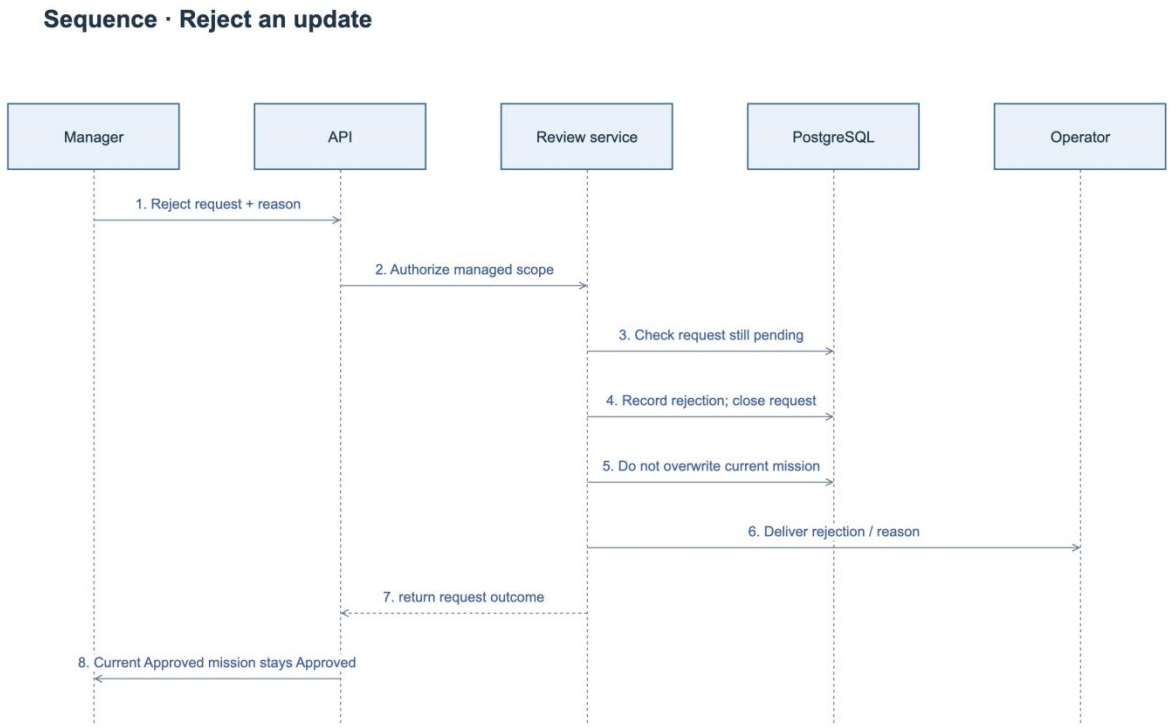


Figure 9. Reject Mission Update Sequence Diagram.

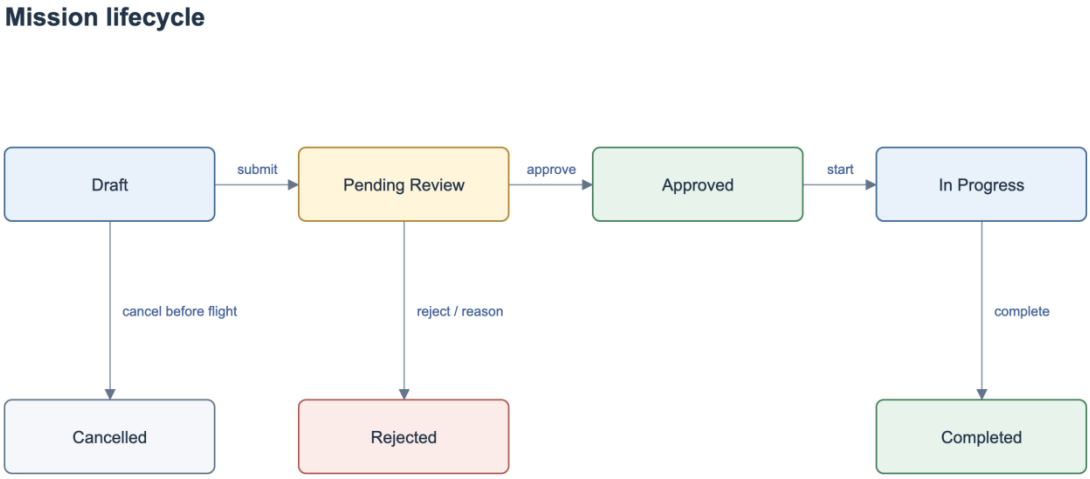


Figure 10. Mission State Diagram.

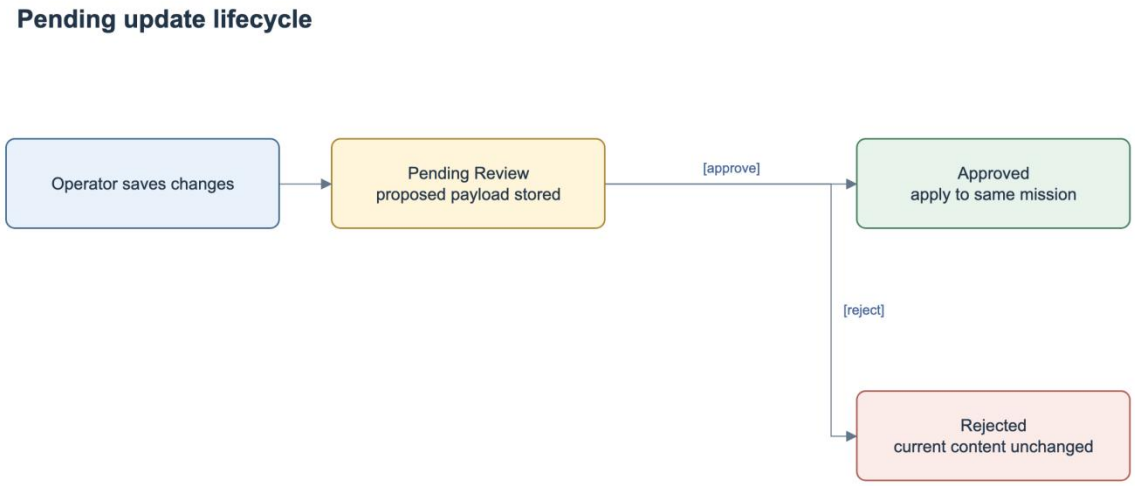


Figure 11. Mission Update Request State Diagram.

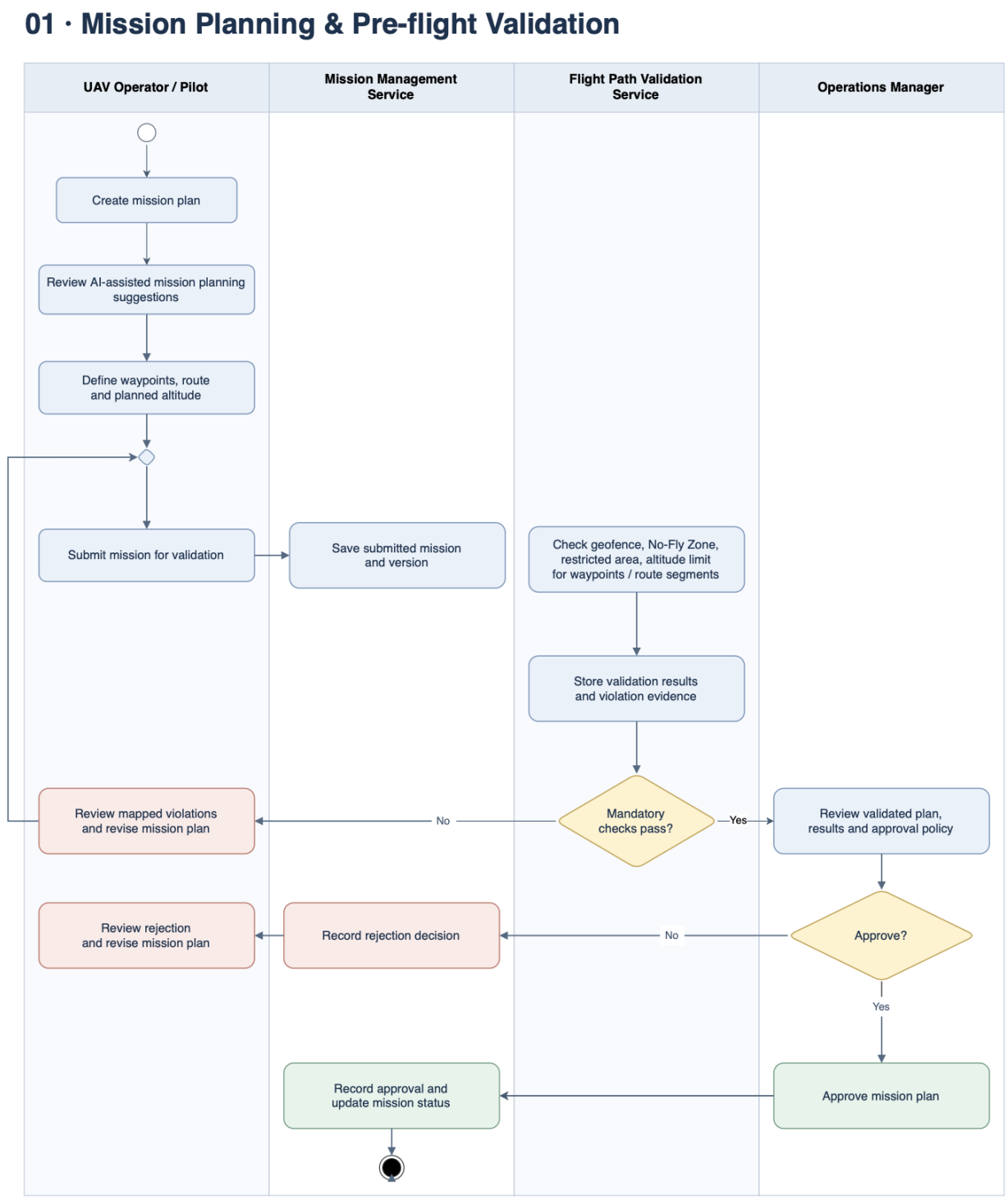


Figure 12. Mission Planning and Pre-flight Validation Workflow.

3.2 Realtime Monitoring and Incident Management

This feature receives timestamped telemetry, maintains the latest authorized mission state and records incident evidence and handling decisions.

3.2.1 Class Diagram

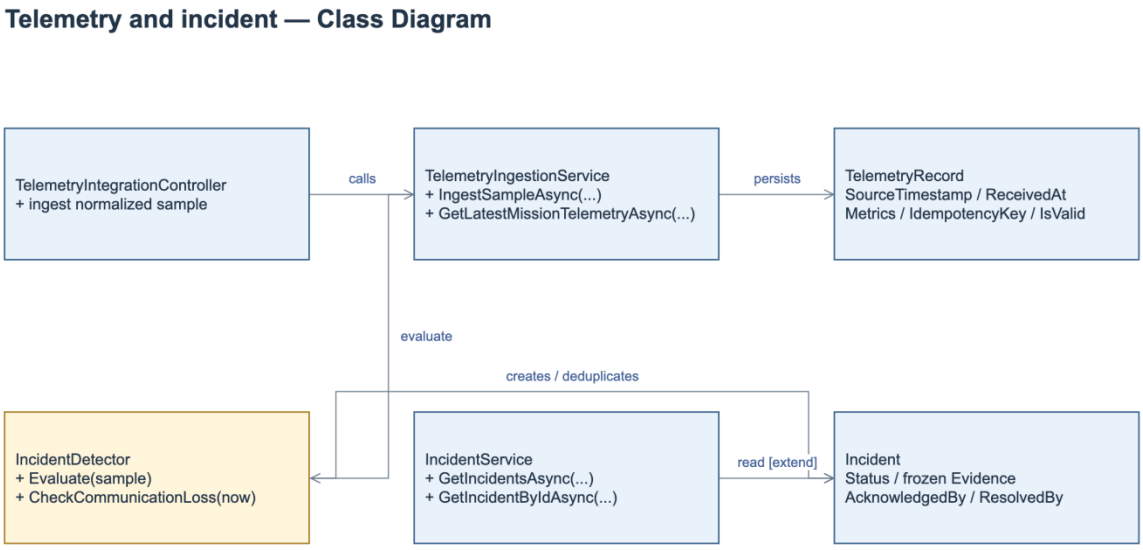


Figure 13. Telemetry and Incident Class Diagram.

Class Specifications

No	Class / Operation	Specification
1	TelemetryIngestion Service.IngestSampl eAsync	Check integration authentication upstream, idempotency, UAV/mission association and source/received timestamps. Persist valid/invalid classification; avoid replacing latest by older out-of-order data.
2	Latest / SSE	Authorize scope, send latest snapshot and incident feed; FE shows unavailable/null/stale. Current sequence IDs reset per connection; durable replay is not promised. Reconnect resynchronizes snapshot and alerts.
3	IncidentDetector / timer	Evaluate battery, altitude and speed against configured unit-aware rules. An independent timer checks communication loss. Rule recovery conditions determine when a new abnormal episode may create another incident.
4	Acknowledge / resolve	Enforce current state, actor and Manager-only resolution with reason; persist frozen source evidence and append-only timeline. Reject conflict, duplicate or out-of-scope mutation without changing evidence.

3.2.2 Ingest and Monitor Telemetry

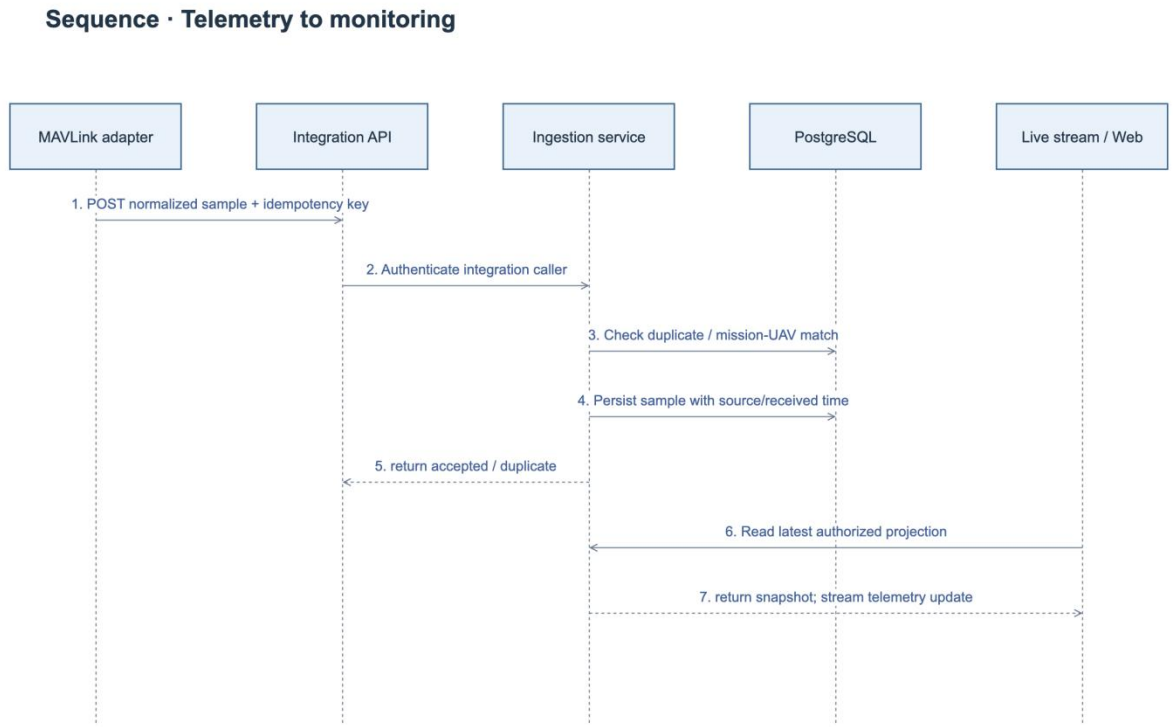


Figure 14. Telemetry Ingestion Sequence Diagram.

3.2.3 Acknowledge and Resolve an Incident

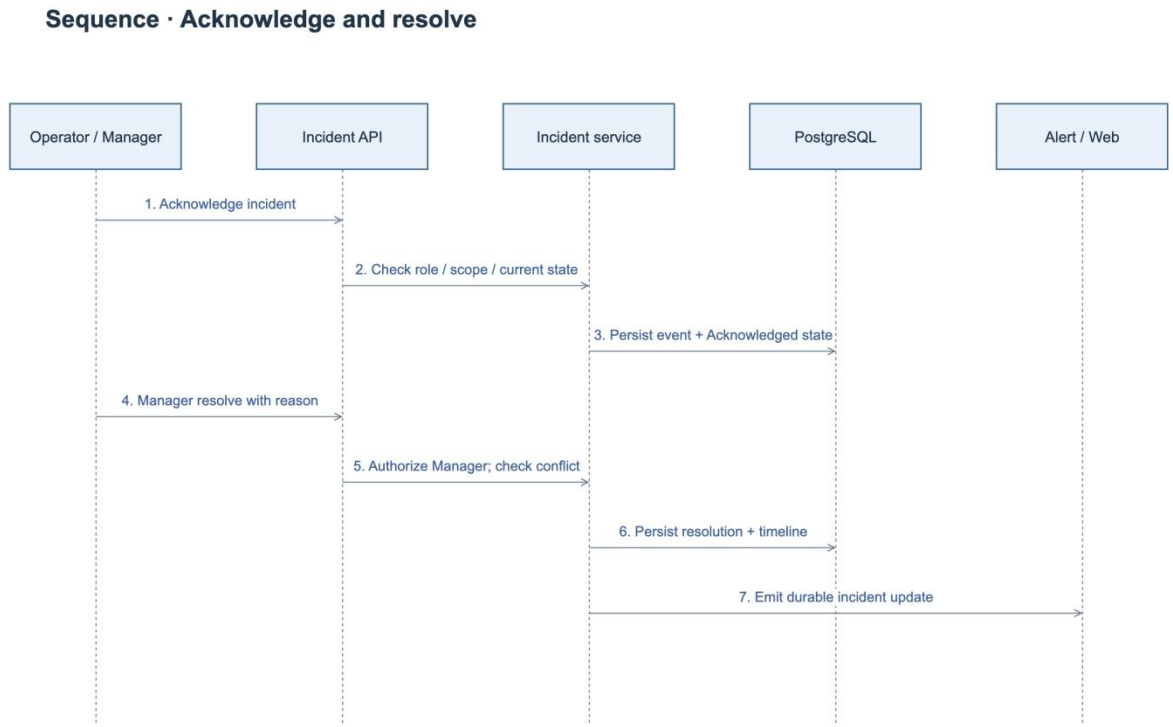


Figure 15. Incident Handling Sequence Diagram.



Figure 16. Incident State Diagram.

3.3 Post-flight History and Reporting

PostFlightService provides history queries, report generation, telemetry playback, mission incidents and operational analytics. Aggregation uses authorized persisted samples and UTC coverage windows. Missing metrics remain null. Mission completion follows the external execution event; it is not a browser flight-control command.

3.3.1 Class Diagram

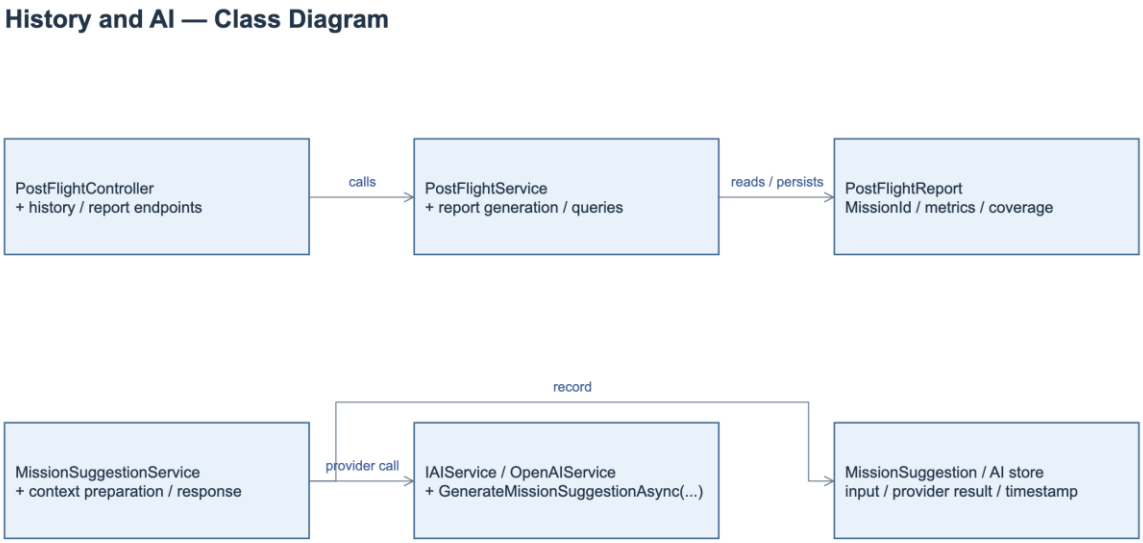


Figure 17. History, Reporting and AI Class Diagram.

Class Specifications

No	Class / Operation	Specification
1	PostFlightService.GetFlightHistoryAsync	Queries scoped mission history with date, UAV and status filters and pagination.
2	PostFlightService.GetReportByMissionIdAsync	Returns the persisted report, source mission, coverage and metrics; missing measurements remain unavailable.
3	PostFlightService.GetTelemetryPlaybackAsync	Returns authorized ordered telemetry positions and timestamps for recorded-route playback.
4	PostFlightService.FinalizeAndGenerateReportAsync	Processes the authorized completion context and aggregates persisted telemetry and incidents into the mission report.

3.3.2 Retrieve a Post-flight Report

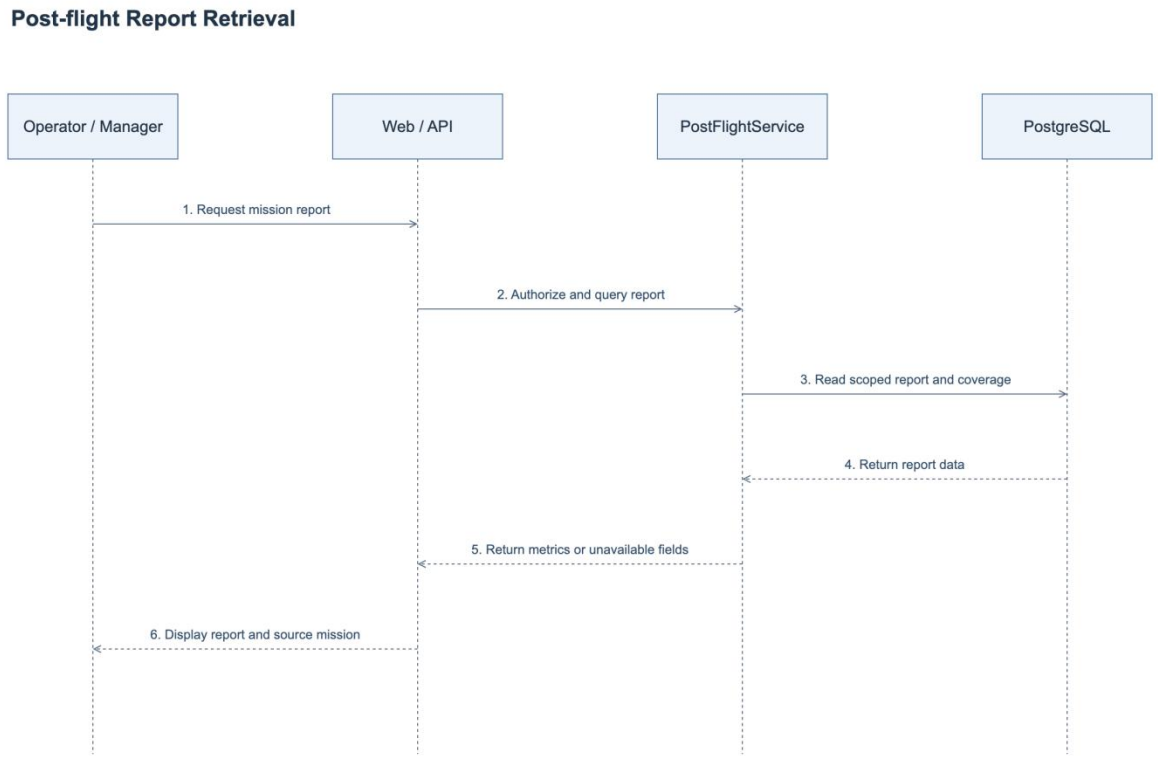


Figure 18. Post-flight Report Retrieval Sequence Diagram.

3.4 AI-assisted Mission Assessment

IAIService.GenerateMissionSuggestionAsync and OpenAIService provide the external AI adapter. MissionSuggestionService prepares and persists mission or update-request context, provider identity, timestamp and assessment outcome. Timeout, rate-limit and invalid-response handling return an unavailable result without changing mission approval. AI output remains advisory.

3.4.1 Class Diagram

The AI assessment classes and their relationships are included in Figure 18. Post-flight reporting and AI assessment share mission and operational evidence; their service responsibilities are specified separately below.

Class Specifications

No	Class / Operation	Specification
1	MissionSuggestionService	Authorizes and prepares mission or update context, calls the configured adapter and persists the assessment outcome.
2	IAIService.GenerateMissionSuggestionAsync	Defines the provider contract for an advisory mission assessment.

3	OpenAIService	Sends bounded authorized context and handles provider timeout, rate limit and invalid responses.

3.4.2 Request an AI Assessment

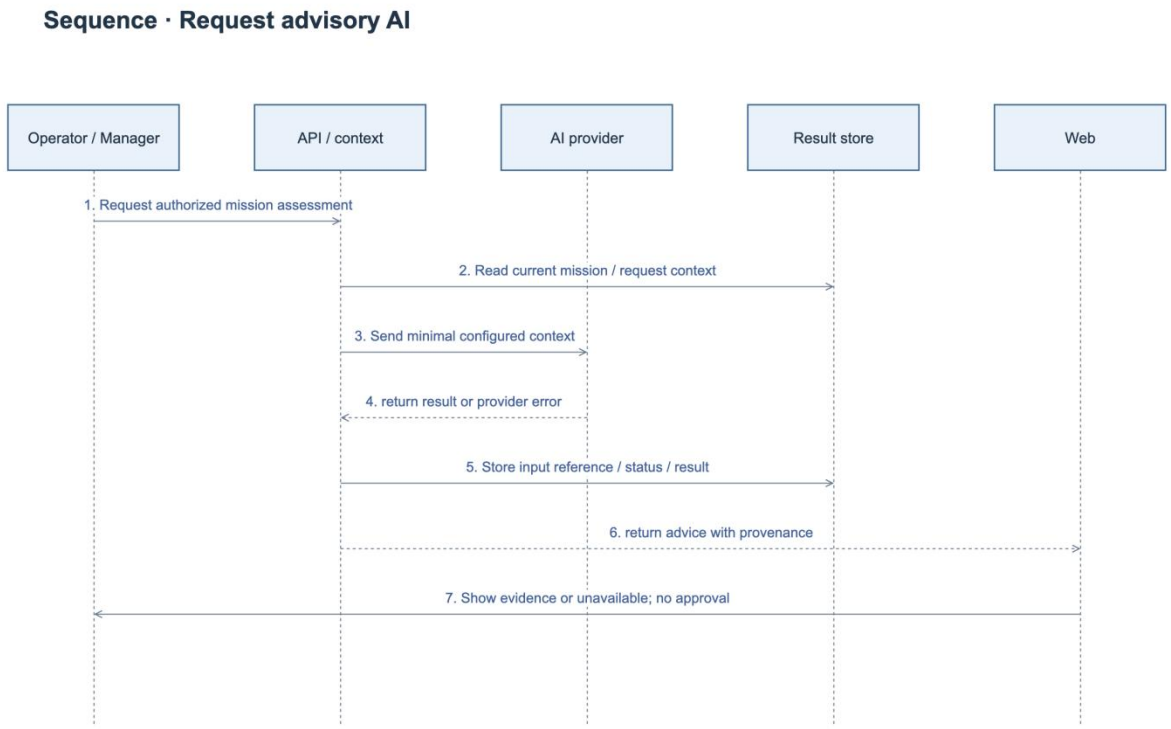


Figure 19. AI Assessment Sequence Diagram.

Automatic flight preparation and workflow ownership

Current ownership: MF1 / WF-01 — NghiaHQ (Ha Quang Nghia), planning and preflight; MF2 / WF-02 — ThongLQ (Le Quoc Thong), realtime monitoring, incidents and flight preparation; MF3 / WF-03 — DucDDA (Dang Dao Anh Duc), post-flight; MF4 / WF-04 — KhanhGNG (Giang Nguyen Gia Khanh), AI assessment. ThongLQ remains team leader and dossier coordinator.

Implemented scope

AutomaticFlightRequested is optional and defaults to false. Changes follow the mission proposal and approval workflow. This release prepares readiness; dispatchEnabled is false. Neither confirmation nor gateway enrollment uploads a mission, arms motors, sends TAKEOFF/AUTO/MISSION_START, or changes the mission to InProgress.

Confirmation conditions

For an approved, opted-in mission with a schedule, the backend opens confirmation at T minus 5 minutes and closes it at the scheduled start. The authenticated operator must keep a visible browser tab online. Presence is sent every 10 seconds and expires after 30 seconds. The selected, unrevoked mission connection must match the active UAV and flight-controller UID, with gateway heartbeat within 30 seconds and telemetry within 15 seconds.

Readiness evidence

Confirmation reruns preflight and binds readiness to the exact plan hash and connection ID. Readiness requires a passing preflight for that hash from the last minute. Changed plan/connection, stale presence or missing telemetry invalidates the conditions; the operator must review and confirm again. Dismissal, a hidden/offline tab or a missed window is not flight consent.

Future execution scope

Autopilot execution remains a separate development phase: firmware/airframe capability detection, mission compilation and upload/read-back, command IDs/deadlines and replay protection, scheduled condition rechecks, local pre-arm/failsafe enforcement, and ACK plus observed flight-state verification. Validate ArduPilot/PX4 SITL, propeller-free bench operation and supervised flight before enabling dispatch. AI advice cannot authorize launch.

Service and API boundary

MissionAutomationService and MissionAutomationController provide GET /api/missions/automation/due and GET /api/missions/{id}/automation, POST presence/confirm/cancel and GET plan below that mission automation route. Owner authorization and uav_operator role apply. MissionAutomationState retains presence/confirmation, confirmed plan hash and connection ID; trusted enrollment and mission leases remain separate. FlightPreparationPrompt polls only when visible/online. Migration: 20261104115300_gateway_enrollment_and_flight_preparation.sql; deploy backend, frontend and gateway package together.

Diagram maintenance

Existing diagrams are unchanged. Update architecture, ERD, mission state and planning/gateway sequences for enrollment and readiness. Confirmation must not transition the mission to InProgress.